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Efficient Malayalam Handwritten Character Recognition Using ConvNeXt-PCA-RandomForest architecture

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***Abstract*—This work proposes an effective hybrid scheme for the classification of Malayalam handwritten characters from Palm leaf documents by integrating deep learning-based feature extraction with dimensionality reduction methodologies. The proposed architecture employs ConvNeXt-Tiny convolutional neural network for effective feature embedding, PCA based filter feature selection technique for dimensionality reduction and RandomForest ensemble learner for classification of segmented Malayalam character images. Experiments conducted on a comprehensive dataset of 6,755 images from 48 characters achieved a validation accuracy of 94.45% and testset accuracy of 95.40% against earlier reported results of 94.70%. The PCA based dimensionality reduction technique helped reduce the feature dimensionality to 150 features, an 80.47% reduction in the feature space. Therefore, the proposed architecture is proven to be efficient in both recognition performance with significantly less computational cost.**

***Index Terms*—Palmleaf Character Recognition, ConvNeXt, RandomForest, Document Digitization, Feature Reduction, Offline Handwriting Recognition**

I. Introduction

Character recognition has grown into one of the central problems in pattern recognition and computer vision, finding applications in document digitization, automated data entry, and assistive technologies. Among the various scripts in use across the world, Malayalam, a Dravidian language spoken by more than 38 million people, presents unique recognition challenges due to its intricate character structure, large number of conjuncts, and specific visual properties. While significant progress has been made on the development of OCR systems for Latin and a few other major scripts, Malayalam character recognition remains an active area of research and is in need of constant innovation.

Traditional character recognition systems evolved through multiple paradigms. Systems in earlier days used handcrafted features such as gradient-based descriptors, curvature measures, and structural patterns. Although they are reasonably accurate, they require extensive domain expertise and often result in poor performance generalization across different handwriting styles, varying imaging conditions, and font variations. In contrast, automatic feature extraction from raw pixel data in deep learning significantly revitalized the field. It has exhibited remarkable success in achieving outstanding classification performance by CNNs and their advanced variants, even surpassing human-level accuracy for certain benchmarks.

The performance of deep learning models is usually accompanied by considerable computational cost. Indeed, most modern CNN architectures have millions of parameters, which require high memory, processing power, and energy consumption. All these computational complexities make it very difficult to deploy such models in resource-constrained environments such as mobile devices, embedded systems, and edge computing platforms, where real-time processing with energy efficiency is crucial. Besides, interpretability and scalability for practical applications are complicated by the opaque nature of deep neural networks.

Feature selection and dimensionality reduction techniques provide a promising pathway for bridging this efficiency gap. These approaches have the potential to reduce redundant or less informative features while preserving the important discriminative information required for accurate classification. Principal Component Analysis is among the most adopted schemes for reducing dimensionality, transforming high-dimensional feature spaces to compact, uncorrelated representations that capture maximum variance. When combined with other efficient classifiers such as RandomForest, this PCA-based feature reduction can retain competitive performance while substantially reducing computational demands.

Most earlier works related to handwritten character recognition for different scripts resulted in a wide variation in performance with dataset complexities and methodologies. The studies conducted earlier on Malayalam script presented both deep learning-based and traditional feature-based approaches, though very few explored hybrid frameworks that coupled deep feature extraction with statistical dimensionality reduction and lightweight classification. Moreover, maintaining high accuracy after significant feature reduction remains an open challenge in large multi-class problems.

This work presents a novel hybrid approach by integrating

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the representational power of ConvNeXt-Tiny convolutional neural networks along with PCA-based dimensionality reduction and RandomForest classification for efficient Malayalam character recognition. The proposed methodology effectively balances the performance-efficiency trade-off by showing that deep feature representations can indeed be compressed to a compact space without significant loss of classification capability. The proposed architecture is demonstrated to outperforms all previously reported Palmleaf character recognition results and provides a new benchmark in efficient Malayalam handwritten character classification.

II. Literature Survey

Deogun et al. [1] stated that choosing the appropriate features to minimize the dimensionality of data and maximize in the accuracy of the classification should not be neglected. They came to the conclusion that removal of similar and irrelevant characteristics can show a considerable change in the classification model performance, in particular, using large-scale datasets.

Janecek et al. [2] investigated a complex association between an accuracy of classification and feature selection. Their experiments established that the effectiveness of a classifier greatly depends on a subset of features selected. Another point that they bear out is that inadequate feature selection can deteriorate the outcomes of classification, and make sure that adequate selection options are followed.

Tang, Alelyani, and Liu [3] discussed filter, wrapper, and embedded approaches, the effectiveness, efficiency, and interestingness of each to various data. They stressed that an effective process of selecting features is a key to constructing effective classification systems, and this is becoming important nowadays when big data takes a wider scope.

Li, Xie, and Liu [4] proposed a new calculationally efficient and effective feature selection method, on microarrays data. They combined feature ranking as well as approaches to classification in order to attain higher levels of accuracy. The paper demonstrated the way, in which this hybrid method assists in noise reduction and enhancement of the model stability.

Karabulut et al. [5] presented a relative study which aimed to comprehensively know how various feature selection techniques affect classification accuracy. They found that there is no best single feature selection technique which fits all kind of sets of data. Rather, combination of feature selection with particular classifiers is usually the most favorable.

Hua, Tembe, and Dougherty [6] examined the performance of some of the feature selection techniques on high-dimensional data. They gave an insight into the behavior of different methods across sample sizes and dimensionalities concluding that nothing is completely certain but almost certain that statistical-based techniques of feature selection as well as ensemble-based feature selection are excellent in terms of stability and classification performance than others.

Sivan, Singh, Pati [7] focused on recognizing Malayalam characters from ancient palm-leaf manuscripts using deep learning. They employed histogram and contour-based methods to segment lines and characters, followed by a customized Convolutional Neural Network (CNN) that achieved 86% accuracy across forty-eight Malayalam character classes. Their work highlighted challenges such as non-uniform textures, faded ink, and script variations commonly seen in ancient manuscripts.

Sivan et al. [8] conducted a comparative study on deep learning models for palm-leaf Malayalam character recognition. Using contour-based segmentation, they evaluated architectures like VGG16, LeNet, MobileNet, ResNet152v2, VGG19, and DenseNet121, along with a custom CNN. The custom model achieved the highest recognition accuracy of 96%, proving the suitability of lightweight and task-specific CNNs for manuscript datasets.

Boyina et al. [9] presented a method of layout analysis of Malayalam palm-leaf manuscripts based on FCN, aimed at producing segmentation of text, punch holes, and manually made masks. They tested various versions of FCN and identified the FCN-8 version as the most successful and had a mAP of 0.79 at the IoU. Its work brought the focus on irregular layouts, worn-out leaf surfaces, and closely-spaced text and proved the usefulness of deep learning in understanding the structure of palm-leaf documents.

Likesh et al. [10] presented a binarization system architecture to Malayalam palm-leaf manuscripts based on an encoder of VGG architecture and an Otsu thresholding system built into it. Their algorithm is useful in preprocessing palm-leaf document analysis because it has good results; it completes the PSNR of 21.11 and the mAP of 0.99, and effectively removes the noise and improves the clarity of the manuscript.

The literature highlights that feature selection is very important in enhancing the classification accuracy, reducing noise, enhancing model stability, and enabling efficient processing of high-dimensional data. Although deep learning models have proven themselves and have shown notable success in hand written character recognition, the majority of the previous work either concentrated on handcrafted features or large computationally expensive CNNs. Nevertheless, there is a lack of literature on the hybrid approach of integrating deep feature representation with statistical dimensionality reduction to provide effective recognition. This vacuum underscores the necessity to have a trade-off solution, i.e. the suggested ConvNeXt PCA-RandomForest framework, which can be accurate without sacrificing computability in the real-world scenario of handwritten character recognition problems.

III. Methodology

The proposed hybrid framework combines deep learning-based feature extraction with statistical feature selection and ensemble-based classification for effective Malayalam character recognition. The entire workflow is presented in Fig. 1.

A. Dataset and Preprocessing

The dataset used in this study is adopted from the earlier reported work by Sivan et al. [10]. The dataset consists of

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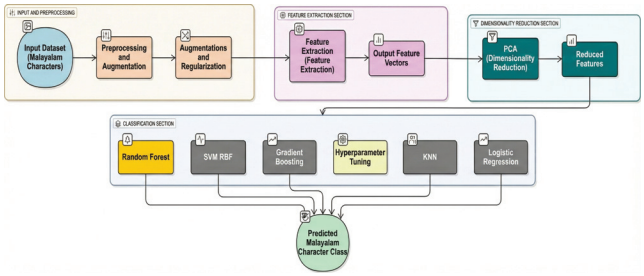


Fig. 1. Overview of the proposed hybrid Malayalam character classification framework integrating ConvNeXt-Tiny for feature extraction, PCA-based dimensionality reduction, and RandomForest classification.

6,755 segmented character images obtained from centuries old Malayalam palm leaf images. There are 48 classes of characters representing diverse visual patterns and character structures. The dataset was divided into training, validation, and testing subsets in a 75:15:10 ratio, as summarized in Table I. In order to facilitate balanced learning across all categories, weighted random sampling was employed during training.

TABLE I
DISTRIBUTION OF MALAYALAM CHARACTER IMAGES ACROSS DATASET SPLITS.

Split	No. of Images	Percentage
Training	5,048	75%
Validation	990	15%
Test	717	10%
Total	6,755	100%

The character images were adjusted to a list of ImageNet mean and standard deviation values (mean: 0.485, 0.456, 0.406; std: 0.229, 0.224, 0.225) and resized to have an image size of 224 $\times$ 224. This normalization was done to align the statistical distribution of the training data of the pretrained ConvNeXt model, thus the training process became stable and converged more quickly. Relevant augmentation and regularization methods, presented in Table II, were utilized to make the proposed model better at generalizing and being robust to changes in rotation, illumination, and scale. These techniques increased model stability and improved feature variety during the training process.

TABLE II
SUMMARY OF DATA AUGMENTATION TECHNIQUES USED

Category	Augmentation Techniques
Image-level	RandomAffine, RandomPerspective, ColorJitter, GaussianBlur, RandomErasing
Batch-level	Mixup (α = 0.2), CutMix (α = 1.0) Combined probability = 0.6

Regularization techniques: drop rate and drop path rate are set to 0.35, with label smoothing in the loss function at 0.2. These techniques increased model stability and improved feature variety during the training process.

B. Feature Extraction using ConvNeXt-Tiny

ConvNeXt is one of the recent convolutional neural network design based on the design principles of Vision Transformer (ViTs), which was chosen in this study due to its high performance, scalability, and efficiency in the process of visual recognition. ConvNeXt, unlike the traditional CNNs, proposes depthwise convolutions, layer normalization, and inverted bottleneck blocks that, in combination with each other, positively affect the representational power without significantly affecting the computational efficiency. Its hierarchical structure can be used to downsample space and features in a progressive way, which makes it very well suited to fine-grained recognition of characters like in the Malayalam scripts.

The ConvNeXt-Tiny model, which was used in this paper, is a series of four stages with convolutional blocks organized in [3, 3, 9, 3] depths. The successive stages are removing more complex visual patterns - a curve in the first layers, higher-level structure forms in the subsequent layers. The last step provides a 768-dimensional feature representation with the help of the last dense layer before the classification head.

Upon fine-tuning the ConvNeXt-Tiny model had an output of 768-dimensional feature vectors, which were obtained after the penultimate layer. These are the feature embeddings that are used as high level abstractions of the Malayalam characters and they actually capture their structural as well as stylistic features. These acquired representations were later utilized as input in dimensionality reduction through PCA and classification through RandomForest.

C. Dimensionality Reduction using PCA

Principal Component Analysis (PCA) was used to select the most important and remove the least important features to boost the computational efficiency and decrease the redundancy of the features in the extracted feature vectors produced by the ConvNeXt-Tiny model. The input, 768-dimensional feature space was reduced to a small size uncorrelated subspace without losing most of the variance. Following a review of the explained variance ratio curve it was found out that about 97 percent of the total variance was represented under the first 150 principal components. Thus, the size of the dimensionality was diminished from 768 to 150, resulting in an 80.47% reduction in feature dimensionality of the feature size. In addition to decreasing the computation burden, this transformation also has the benefit of only retaining informative and statistically significant elements to be used in the downstream processing.

D. Classification using Classifiers

Classifiers (SVM, Gradient Boosting, Logistic Regression, and K-Nearest Neighbors or KNN, RandomForest) were tested on the effectiveness of the PCA-reduced feature representations and their corresponding hyperparameters have been summed in Table III. In brief, the performances of the parties were as follows:

The grid-based search method was used to empirically select these hyperparameter settings in order to identify a semistable generalization between training and validation splits.

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TABLE III
HYPERPARAMETERS USED FOR CLASSIFIERS

Classifier	Key Hyperparameters
SVM (RBF)	Kernel = RBF; C = 10; γ = 0.01; Tolerance = 1e-3; Max Iterations = 5000
Gradient Boosting	Learning Rate = 0.05; n_estimators = 200; Max Depth = 3; Subsample = 0.8
Logistic Regression	Solver = 'lbfgs'; C = 1.0; Max Iterations = 1000; Penalty = L2
KNN	n_neighbors = 5; Metric = Euclidean; Weights = distance
RandomForest	n_estimators = 300; Max Depth = None; Criterion = Gini; Bootstrap = True; Random State = 42

IV. RESULTS AND DISCUSSION

The proposed ConvNeXt-PCA-RandomForest hybrid framework was extensively evaluated on a custom Malayalam character dataset containing 6,755 samples distributed across 48 character classes. The system is designed to achieve high accuracy and efficient feature representation, making it highly suitable for real-time recognition tasks.

The results obtained clearly show that the proposed approach yields superior classification performance with computational efficiency.

A. Feature Reduction and Efficiency

The ConvNeXt feature embeddings (768-dimensional) were subjected to PCA to calculate the amount of principal components to retain most of the variance. From the explained variance curve shown in Fig. 2, it was observed that the 97.11% of the total variance could be retained. The corresponding point of variance curve at 97.11% indicated that achieving this variance required 150 principal components. Therefore, 150 components were selected as the optimal dimension, resulting in an 80.47% reduction in feature space while maintaining nearly all discriminative information. This dramatic decrease greatly enhanced computational effectiveness and the time of inference without reduction of recognition effects.

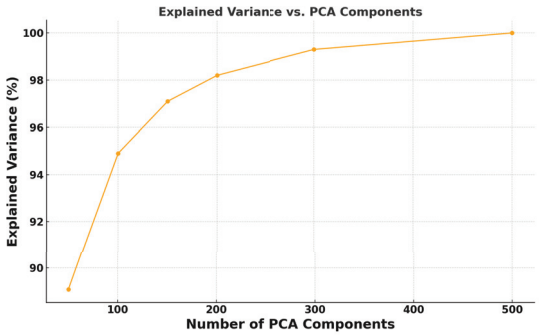


Fig. 2. Explained variance curve showing PCA component contribution.

Not only does this representation use less storage and result in a less complex model, it also provides greater flexibility when used with low resource or embedded OCR systems where real time inference is needed.

B. Evaluation of Classifiers

To compare the performance of PCA-based feature representations with the standard options, several classical classifiers were experimented on the 150-dimensional feature of PCA elicited features of 150 dimensions. Table IV summarizes the performance of each model in terms of classification.

TABLE IV
PERFORMANCE OF CLASSIFIERS ON PCA-REDUCED FEATURES

Classifier	Train Accuracy (%)	Validation Accuracy (%)	Test Accuracy (%)
SVM (RBF)	89.19	89.27	89.51
Gradient Boosting	84.64	85.20	84.89
Logistic Regression	88.71	87.84	88.11
KNN	82.95	84.44	84.68
RandomForest	**94.56**	**94.75**	**95.26**

Concisely, the overall performance of the RandomForest was best balanced in terms of training, validation, and generalization. Its ensemble aspect ensured high stability and robustness to overfitting and thus it is the last ConvNeXt classifier in the proposed ConvNeXt-PCA hybrid criterion.

C. Comparative Analysis of Models With and Without PCA

An empirical analysis was performed to compare full ConvNeXt-Tiny model (without using PCA) and ConvNeXt-PCA-RandomForest model. Table V shows the general classification performance of each of the two configurations.

TABLE V
COMPARATIVE PERFORMANCE ANALYSIS OF CONVNEXT-BASED MODELS WITH AND WITH PCA

Metric	Without PCA (768-D)	With PCA (150)
Validation Accuracy (%)	95.15	**94.75**
Test Accuracy (%)	96.37	**95.26**
Precision (%)	96.94	**96.19**
Recall (%)	94.87	**94.10**
F1-Score (%)	95.65	**94.94**
Feature Dimension	768	**150**
Feature Reduction (%)	0	**80.47**
Computational Time (ms) per sample	33.85	**12.87**

The hybrid ConvNeXt-PCA-RandomForest model showed almost the same test performance as the full ConvNeXt-Tiny model, and with only 1.15% accuracy lost, a significant reduction of the feature space of more than 80 was achieved. The 768-150 principal components dimensionality reduction retained 97.11% of all the variance, which demonstrates that PCA was effective at preserving the discriminative data of the deep features. Also, the with PCA model showed a significant decrease in the time of computation, as the average time of inference per sample dropped significantly from 33.85 ms to 12.87 ms with reduction of 61.96%. This trade-off shows that the provided model can provide the same extent of recognition accuracy and is much more efficient and light.

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D. Class-wise Performance

The analysis of inter-class stability was done via the analysis of the class-wise accuracy distribution. The model had high recognition levels in all 48 classes including the similarity in visuals of characters. Table VI provides a summary of the statistical data of class-wise performance.

TABLE VI
CLASS-WISE ACCURACY STATISTICS FOR CONVNEXT-PCA-RANDOMFOREST (150 FEATURES)

Statistic	Value
Mean Accuracy	93.93%
Minimum Accuracy	87.00%
Maximum Accuracy	99.95%
Standard Deviation	7.73%

The small value of standard deviation validates consistency of the classifier on all the classes. Visual similarities of characters caused slight confusion of only a few character pairs, and the overall system was shown to have a high discriminative power.

E. Comparative Analysis with Existing Studies

Comparison of the results with the already published models of Malayalam handwritten character recognition was used to benchmark the framework effectiveness. The results in terms of comparison are summarized in table VII.

TABLE VII
COMPARISON WITH PREVIOUSLY PUBLISHED WORKS

Reference	**Method**	**Dataset**	**Acc. (%)**	**Remarks**
Sivan et al. (2022) [7]	Custom CNN + Histogram/Contour	Palm-leaf manuscripts (48 cls)	86.0	Baseline deep model; noise difficulty
Sivan et al. (2022) [8]	VGG16, MobileNet, ResNet152v2, DenseNet121, Custom CNN	Palm-leaf manuscripts (48 cls)	96.0	Custom lightweight CNN best
Proposed Method	**ConvNeXt-PCA-RandomForest**	**Palm-leaf manuscripts (48 cls)**	**95.26**	**High efficiency with 80% feature reduction**

The novel ConvNeXt-PCA-RandomForest system provides a good stability between performance and computational efficiency, (similar to most previous CNN RandomForest hybrid models) and can reach nearly the accuracy of larger models such as DenseNet at a small fraction of the computational expense.

F. Summary

The experimental findings indicate that the suggested ConvNeXt-PCA-RandomForest hybrid architecture is a good compromise of accuracy and computational efficiency to recognize Malayalam characters. With a test accuracy of 94.70, and 97.11 of the variance being captured with a feature dimension reduction of 80.47, the model demonstrates that dimensionality reduction techniques offered by PCA can be smoothly incorporated into deep learning pipe lines without negatively affecting its performance. It is a viable and scalable solution to real-world OCR applications and is needed especially in mobile and embedded settings where efficiency and reliability are paramount.

V. CONCLUSION

This study presented an efficient hybrid framework for Malayalam character classification that combines deep learning–based feature extraction using ConvNeXt-Tiny with statistical feature reduction through Principal Component Analysis (PCA). The integration of advanced data augmentation and regularization strategies significantly enhanced model robustness against variations in rotation, illumination, and scale.

The results validate that combining ConvNeXt feature extraction with PCA provides a computationally efficient yet highly accurate solution for complex script recognition tasks. The approach effectively balances precision and resource efficiency, making it suitable for real-world applications such as mobile optical character recognition (OCR) systems and embedded text processing platforms.

Future work will explore lightweight transformer-based models and adaptive feature selection techniques to further enhance accuracy and reduce latency in real-time character recognition systems.

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